

Vanishing Elasticity Limit and Defect Analysis in the Landau-de Gennes Model

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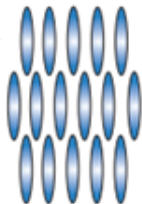
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- 2 Motivations and Main Results
- 3 Sketch of the Proof

Liquid Crystals

Liquid crystals (LCs) are anisotropic fluids. The anisotropy arises from the directional nature of the molecular geometry, physical, or chemical properties.

LCs are between anisotropic crystalline and isotropic liquid phases.



crystalline

Order



isotropic

Disorder

v.s.

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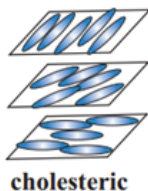
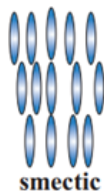
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Let $\mathbf{Q} \in \mathbb{S}_0$ and $\lambda_1, \lambda_2, \lambda_3$ be the eigenvalues of $\mathbf{Q}$. Then

$$\mathbf{Q} = \lambda_1 \mathbf{n}_1 \otimes \mathbf{n}_1 + \lambda_2 \mathbf{n}_2 \otimes \mathbf{n}_2 + \lambda_3 \mathbf{n}_3 \otimes \mathbf{n}_3,$$

where $\mathbf{n}_1, \mathbf{n}_2, \mathbf{n}_3 \in \mathbb{S}^2$ with $\mathbf{n}_i \cdot \mathbf{n}_j = \delta_{ij}$ ($1 \leq i, j \leq 3$). Rewrite $\mathbf{Q}$ as

$$\mathbf{Q} = s \left(\mathbf{n} \otimes \mathbf{n} - \frac{1}{3} \mathbf{I} \right) + r \left(\mathbf{m} \otimes \mathbf{m} - \frac{1}{3} \mathbf{I} \right), \quad s, r \in \mathbb{R},$$

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- ❶ **Isotropic:** If $s = r = 0$, namely, $\mathbf{Q} = \mathbf{O}$.
- ❷ **Biaxial:** If s and r are different and non-zero.
- ❸ **Uniaxial:** If $s = r \neq 0$ or $s = 0, r \neq 0$ or $s \neq 0, r = 0$. In particular,

$$\mathbf{Q} = t \left(\mathbf{u} \otimes \mathbf{u} - \frac{1}{3} \mathbf{I} \right), \quad \mathbf{u} \in \mathbb{S}^2.$$

The free energy functional is

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$$f_e(\mathbf{Q}) = \frac{L_1}{2} |\nabla \mathbf{Q}|^2 + \frac{L_2}{2} \partial_j \mathbf{Q}_{ij} \partial_k \mathbf{Q}_{ik} + \frac{L_3}{2} \partial_k \mathbf{Q}_{ij} \partial_j \mathbf{Q}_{ik}, \quad .$$

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Stable equilibrium configurations of the liquid crystalline system with a domain $\Omega \subset \mathbb{R}^d$ ($d = 2, 3$) correspond to local minimizers of $F(\cdot, \Omega)$.

Vanishing Elasticity Limit: Background

Main reductive assumption: Let $L_2 = L_3 = 0$ for simplicity.

Let $\Omega \subset \mathbb{R}^3$ be a bounded domain. Consider the Landau-de Gennes energy

$$E_\varepsilon(\mathbf{Q}, \Omega) := \int_{\Omega} e_\varepsilon(\mathbf{Q}) dx, \quad e_\varepsilon(\mathbf{Q}) := \frac{1}{2} |\nabla \mathbf{Q}|^2 + \frac{1}{\varepsilon^2} f(\mathbf{Q}), \quad (\text{LdG})$$

where

$$f(\mathbf{Q}) = k - \frac{a}{2} \text{tr } \mathbf{Q}^2 - \frac{b}{3} \text{tr } \mathbf{Q}^3 + \frac{c}{4} (\text{tr } \mathbf{Q}^2)^2, \quad \mathbf{Q} \in \mathbb{S}_0.$$

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The observation in physics: Elastic constants are so small compared to the bulk constants.

Question (Vanishing elasticity limit)

The asymptotic behavior of minimizers (or critical points) to (LdG) as $\varepsilon \rightarrow 0^+$.

Limiting Functional

The vacuum manifold is

$$\mathcal{N} := \left\{ s_* \left(\mathbf{n} \otimes \mathbf{n} - \frac{1}{3} \mathbf{I} \right) : \mathbf{n} \in \mathbb{S}^2 \right\} = f^{-1}(0),$$

where

$$s_* := s_*(a, b, c) = \frac{b + \sqrt{b^2 + 24ac}}{4c}.$$

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The limiting energy functional is

$$E(\mathbf{Q}, \Omega) := \frac{1}{2} \int_{\Omega} |\nabla \mathbf{Q}|^2 dx, \quad \mathbf{Q} \in H^1(\Omega, \mathcal{N}). \quad (\text{Dir})$$

Previous Results: Bounded Energy

Consider the case that $\exists M > 0$, s.t. $\forall \varepsilon \in (0, 1)$,

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$$\mathbf{Q}_\varepsilon = \mathbf{Q}_0 + \varepsilon \mathbf{Q}_0^1 + \varepsilon^2 \mathbf{Q}_0^2 + \dots$$

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- ③ [Contreras-Lamy '22, Feng-Hong '22]: Generalizing results by Nguyen-Zarnescu to the cases with non-zero L_2 and L_3 .

Previous Results: Logarithmic Energy

Assume that $\exists M > 0$, s.t. $\mathbf{Q}_\varepsilon$ minimizes (LdG), and

$$\frac{E_\varepsilon(\mathbf{Q}_\varepsilon, \Omega)}{\log \frac{1}{\varepsilon}} + \|\mathbf{Q}_\varepsilon\|_{L^\infty(\Omega)} \leq M, \quad \forall \varepsilon \in (0, 1).$$

Let $\{\mu_\varepsilon\}_{\varepsilon \in (0,1)} \subset (C_0(\Omega))'$ be Radon measures with

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Up to a subsequence, $\mu_\varepsilon \rightharpoonup^* \mu_0$ in the sense of measures.

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- ① [Canevari '17]: $\mu_0 = \sum_i \frac{\pi s_*^2}{2} \mathcal{H}^1 \llcorner L_i$, where $\{L_i\}$ are segments of lines.
For any $K \subset\subset \Omega \setminus \text{supp}(\mu_0)$,

$$E_\varepsilon(\mathbf{Q}_\varepsilon, K) \leq C(a, b, c, K, M),$$

returning to the previous setting. The idea is from [Lin-Rivière '99] for the Ginzburg-Landau Model.

Previous Results: Logarithmic Energy

Assume that $\exists M > 0$, s.t. $\mathbf{Q}_\varepsilon$ minimizes (LdG) , and

$$\frac{E_\varepsilon(\mathbf{Q}_\varepsilon, \Omega)}{\log \frac{1}{\varepsilon}} + \|\mathbf{Q}_\varepsilon\|_{L^\infty(\Omega)} \leq M, \quad \forall \varepsilon \in (0, 1).$$

Let $\{\mu_\varepsilon\}_{\varepsilon \in (0,1)} \subset (C_0(\Omega))'$ be Radon measures with

$$\mu_\varepsilon(A) := \frac{E_\varepsilon(\mathbf{Q}_\varepsilon, A)}{\log \frac{1}{\varepsilon}}, \quad A \subset \Omega.$$

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- ② [W.-Zhang '24]: Extending results in [Canevari '17] to include sextic bulk energy potentials.

Main Results: Bounded Energy

Theorem (Fu-Wang-W. arXiv: 2507.14955)

Suppose $\{\mathbf{Q}_\varepsilon\}_{\varepsilon \in (0,1)} \subset H_{\text{loc}}^1(\Omega, \mathbb{S}_0)$ are local minimizers of (LdG), s.t.

$$E_\varepsilon(\mathbf{Q}_\varepsilon, \Omega) + \|\mathbf{Q}_\varepsilon\|_{L^\infty(\Omega)} \leq M, \quad \forall \varepsilon \in (0, 1).$$

Then, the following properties hold.

- ① $\forall p \in (1, 3)$, $\{\mathbf{Q}_\varepsilon\}_{\varepsilon \in (0,1)}$ is relatively compact in $W_{\text{loc}}^{1,p}(\Omega, \mathbb{S}_0)$.
- ② $\forall K \subset\subset \Omega$, $\exists C = C(a, b, c, K, M) > 0$, s.t.

$$\int_K \frac{1}{\varepsilon^2} f(\mathbf{Q}_\varepsilon) dx \leq C\varepsilon.$$

Main Results: Logarithmic Energy

Theorem (Fu-Wang-W. arXiv: 2508.01811)

Suppose $\{\mathbf{Q}_\varepsilon\}_{\varepsilon \in (0,1)} \subset H_{\text{loc}}^1(\Omega, \mathbb{S}_0)$ are local minimizers of (LdG), s.t.

$$E_\varepsilon(\mathbf{Q}_\varepsilon, \Omega) \leq M \left(\log \frac{1}{\varepsilon} + 1 \right), \quad \|\mathbf{Q}_\varepsilon\|_{L^\infty(\Omega)} \leq M, \quad \forall \varepsilon \in (0, 1).$$

Then, the following properties hold.

- 1 $\forall p \in (1, 2)$, $\{\mathbf{Q}_\varepsilon\}_{\varepsilon \in (0,1)}$ is relatively compact in $W_{\text{loc}}^{1,p}(\Omega, \mathbb{S}_0)$.
- 2 $\forall K \subset\subset \Omega$, $\exists C = C(a, b, c, K, M) > 0$, s.t.

$$\int_K \frac{1}{\varepsilon^2} f(\mathbf{Q}_\varepsilon) dx \leq C.$$

Remarks on Main Results

- ① Our convergence results in the main theorems are already optimal, because of the analysis in [Brézis-Coron-Lieb '86] and [Canevari '17]. Indeed, we can provide explicit examples of it.

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- 3 The first theorem extends [Majumdar-Zarnescu '10].
- 4 The second theorem generalizes [Bourgain-Brézis-Mironescu '04] for the Ginzburg-Landau model to the Landau-de Gennes model.

Table of Contents

- 1 Brief Introduction of Liquid Crystals
- 2 Motivations and Main Results
- 3 Sketch of the Proof

Characterization of the Bad Behavior

We first introduce the regular scale to give a quantitative characterization of regularity.

Definition (Regular scales)

Let $\mathbf{Q} \in C^\infty(\Omega, \mathbb{S}_0)$. For $x \in \Omega$ and $\Lambda > 0$, define

$$r(\mathbf{Q}, x) := \sup\{r > 0 : r\|(|\nabla \mathbf{Q}| + r|D^2 \mathbf{Q}|)\|_{L^\infty(B_r(x))} \leq 1\},$$
$$r^\Lambda(\mathbf{Q}, x) := \sup\{r > 0 : E_\varepsilon(\mathbf{Q}, B_r(x)) \leq \Lambda r\}.$$

Assume $\mathbf{Q}_\varepsilon \in H^1(\Omega, \mathbb{S}_0)$ is a local minimizer of (LdG). For parameters $\Lambda, r > 0$, we define the type I bad set as

$$\text{Bad}_I(\mathbf{Q}_\varepsilon; r) := \{y \in \Omega : r(\mathbf{Q}_\varepsilon, y) < r\}.$$

Also let the type II bad set of $\mathbf{Q}_\varepsilon$ be

$$\text{Bad}_{II}(\mathbf{Q}_\varepsilon; r, \Lambda) := \{y \in \Omega : r^\Lambda(\mathbf{Q}_\varepsilon, y) < r\}.$$

Difficulties and Methods

- **Step 1.** We establish the volume estimates of r -neighborhoods of bad sets, i.e., under appropriate settings, we intend to control

$$\mathcal{L}^3(B_r(\text{Bad}_{\text{II}}(\mathbf{Q}_\varepsilon; r, \Lambda))), \quad \mathcal{L}^3(B_r(\text{Bad}_{\text{I}}(\mathbf{Q}_\varepsilon; r))).$$

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- **Step 2:** Case with bounded energy: The proof is completed by employing arguments in [Wang-Wei-Wu '25], studying fractional Allen-Cahn equations.

Difficulties and Methods

- **Step 3:** Case of Logarithmic energy bound: Situations are very different and more obscurities occur! Due to the limitation of the analysis, the estimate of

$$\mathcal{L}^3(B_r(\text{Bad}_{\text{II}}(\mathbf{Q}_\varepsilon; r, \Lambda)))$$

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- Outside the bad set with the scale $\varepsilon^{\frac{1}{4}}$, we combine the results with bounded energy.

Thank you for listening!